

Project Title

Resource Coordination Platform for Community Resilience

Background

In many developing regions and urban centers, community-based organizations (CBOs) and informal mutual aid groups struggle to coordinate help during localized crises. While social media groups are used for outreach, they lack the structure to track resource availability, verify needs, and manage volunteer logistics in real-time. Existing disaster management software is often designed for government agencies and is too complex for grassroots community leaders.

Problem Description

Grassroots social initiatives face significant hurdles in matching "haves" with "needs." Information is often fragmented across various chat groups, leading to "donation fatigue" or critical needs going unmet. Without a centralized, lightweight system, small-scale social workers cannot effectively:

- Track real-time inventory of donated goods (food, medicine, tools).
- Verify the status of a request to prevent duplication of efforts.
- Manage a pool of volunteers with specific skills (e.g., first aid, plumbing, transport).
- Ensure data privacy for vulnerable individuals seeking help.

Proposed Solution

This project proposes a **SaaS-based Full-Stack Platform** designed for community resilience. The system provides a centralized dashboard for CBOs to manage logistics and a public-facing portal for community members. Unlike traditional CRMs, this is built for rapid deployment and high-concurrency during peak usage. The platform will feature a **Customizable Workflow Engine** where admins can define "Resource Categories" (e.g., "Emergency Shelter" or "Food Bank") without technical expertise. It will rely on robust database design and real-time state management rather than AI to ensure 100% reliability in high-stakes environments.

Objectives

The main objectives of this assignment are:

- To analyze the logistics and data flow challenges of grassroots mutual aid groups.
- To design a multi-tenant system architecture that allows different communities to host their own private coordination hubs.
- To implement a full-stack web and mobile responsive application for real-time resource tracking.
- To develop a role-based access control (RBAC) system to ensure the security of vulnerable community data.
- To evaluate the system's performance under simulated high-traffic "crisis" scenarios.

Scope of the Project

This project focuses on the "Digital Divide" and "Resource Inequity," specifically helping local communities organize support during localized crises (like natural disasters or economic shifts) without relying on complex enterprise tools.

The project includes the design and implementation of a complete end-to-end web and mobile-responsive ecosystem.

Included in Scope: (can refine as needed)

- Web Application (Admin/Coordinator Interface):
Centralized dashboard for verifying and approving resource requests.
Inventory management system for tracked goods.
Volunteer dispatching and task assignment module.
Visual reporting on "Need vs. Fulfillment" metrics.
- Mobile-Responsive Web Client (User/Volunteer Interface):
Simplified portal to "Request Help" or "Offer Resources."
Real-time notifications for volunteers when a nearby task is assigned.
Offline-first capabilities to allow data entry in low-connectivity areas.
- Backend Services:
Multi-tenant support: Allows multiple organizations to use the same platform with isolated data.
RESTful API: To power both the coordinator dashboard and the public portal.
WebSockets: For real-time updates of resource availability.

Out of Scope:

- Machine learning-based demand forecasting or AI chatbots.
- Integration with IoT hardware or GPS tracking devices.
- Financial transaction processing or legal contract management.

Expected Outcomes

- A functional, high-performance prototype for *collaborative online space/* platform.
- Comprehensive technical documentation including ER-Diagrams, System Architecture, and API specifications.
- A deployment-ready SaaS infrastructure utilizing Docker or similar containerization.
- A final report analyzing the system's ability to handle data integrity during high-volume requests.
- A demonstration using a simulated crisis scenario (e.g., a local flood response).